

Compressible Flow Review

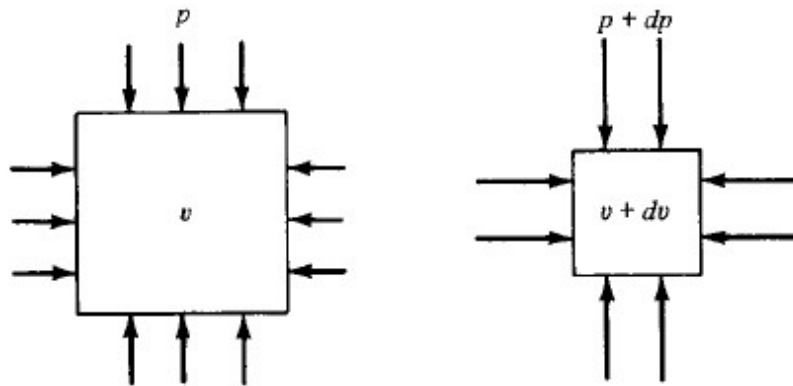
Introduction

- The study of propulsion is intimately linked to the understanding of internal fluid mechanics. This means that duct flows are of prime interest to propulsion.
- The physical phenomena in a real duct include the following effects:
 - Friction
 - Heat transfer through the walls
 - Chemical reaction within the duct
 - Area variation of the duct
 - Compressibility effects, e.g. appearance of shocks

Compressibility

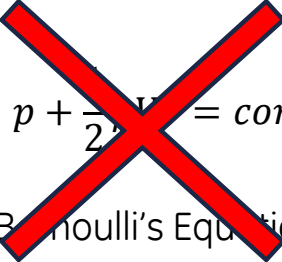
- Physically, the compressibility is the fractional change in volumen of the fluid element per unit change in pressure.

$$\tau = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right) \xrightarrow{\text{yields}} \frac{1}{\rho} \frac{d\rho}{dp} \xrightarrow{\text{yields}} d\rho = \rho \tau dp$$



Compressible Flow

- Compressible flow routinely defined as variable density flow; this in contrast to incompressible flow (liquid and some gases flows under certain conditions) where density changes are negligible and therefore the assumption can be made with reasonable accuracy.
- Hence, the following equation does not hold and should be dismissed from now on


$$p + \frac{\rho V^2}{2} = \text{const}$$

Bernoulli's Equation

Compressible Flow

- A flow must be treated as compressible when the pressure gradients in the flow field are large enough such that, in combination with a large enough value of the compressibility, τ , the resulting density changes are too large to ignore.
- For gases, this occurs when the flow Mach number is greater than about 0.3.

Compressible Flow

- Since the density of the fluid is no longer constant in compressible flows, there are now four dependent variables to be determined throughout the flow field. These are pressure, temperature, density, and flow velocity. Two new variables, temperature and density, have been introduced and two additional equations are required for a complete solution.
- These are the *energy equation* and the fluid *equation of state*. These must be solved simultaneously with the *continuity* and *momentum* equations to determine all the flow field variables.

Ideal Gas

- What is an ideal gas?
- The ideal gas law that relates the pressure, density, and the absolute temperature of the gas may be derived rigorously from the kinetic theory of gases. Prominent assumptions in its formulation are (1) intermolecular forces between the molecules are negligibly small and (2) the volume of molecules that occupy a space is negligibly small and may be ignored.
- For a perfect gas, pressure, density and temperature are related through the following *equation of state*.

$$P = \rho RT$$

- Where R is the specific gas constant, which is a different value for different gases, defined as follows:

$$R = \frac{\bar{R}}{MW}$$

Internal Energy and Enthalpy

- Internal Energy
 - The sum of the translational, rotational, vibrational and electronic energies of a given molecule. And is denoted by the convention E .
- Enthalpy
 - In many thermodynamic analyses the sum of the internal energy E and the product of pressure p and volume V appears. Because the sum $E + pV$ occurs so frequently it is convenient to give the combination a name, **enthalpy**, and a distinct symbol, H .

Since both are properties, this can both be expressed on a unit mass basis

$$h = e + pv$$

For a perfect gas, both e and h are functions of temperature only, related by C_v and C_p , which can behave as constant for a certain range of temperature, thus

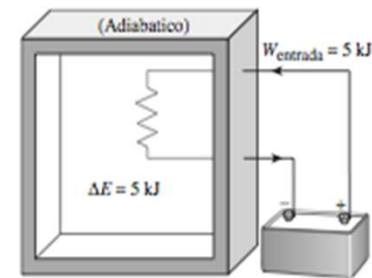
$$\begin{aligned}e &= C_v T \\ h &= C_p T\end{aligned}$$

How are C_p and C_v related?

1st law of thermodynamics

“Energy cannot be created or destroyed but it can be transformed”

- Defines the system's property:
 - Total Energy change (E_T).
 - The difference between two states
- Introduces the idea of energy balance:
 - There cannot be losses!



$$\Delta E_T = -W_{ad}$$

$$E_{\text{state}_2} - E_{\text{state}_1} = -W_{ad_2} + W_{ad_1}$$

General energy balance

Net change in the amount of energy within the system during a process	=	Net amount of energy transferred in the system during a process OR final state	-	Net amount of energy transferred out the system during a process OR initial state
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$$\Delta E_T = Q - W + m$$

1st law of thermodynamics

- For a give de , there are in general an infinite number of different processes by which heat can be added and work done on the system.
- We are primarily concerned with three type of processes:
 1. *Adiabatic process*
 2. *Reversible process*
 3. *Isentropic process*

2nd law of thermodynamics: Entropy

- Entropy (S):
 - Energy lost in the form of heat transfer at the boundary system.
 - Property based on Clausius inequality
 - Disorder or uncertainty of the system
 - Generated by the presence of irreversibilities
- Property of the system:
 - Measure a change in entropy of internally reversible processes
- Irreversibilities:
 - Factors that enable energy loss
 - Friction, expansion of gas/liquids, heat transfer at finite temperature...

$$\oint \left(\frac{\delta Q}{T} \right)_b \leq 0$$

Clausius' inequality

$$\Delta S = S_2 - S_1 = \int_1^2 \left(\frac{\delta Q}{T} \right)_{int rev}$$

Definition of entropy change

Entropy balance

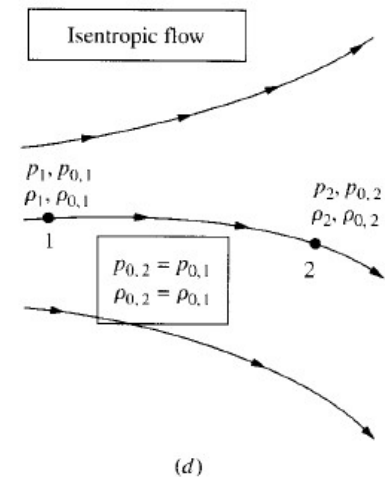
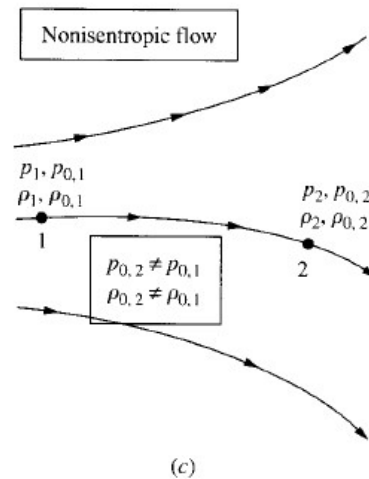
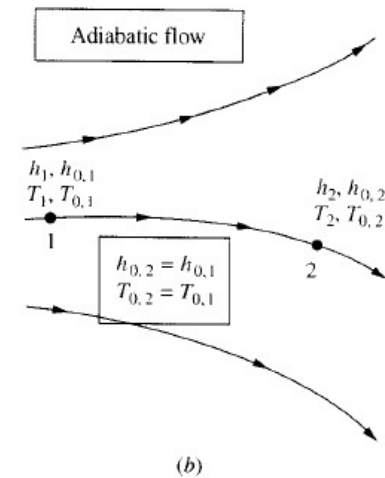
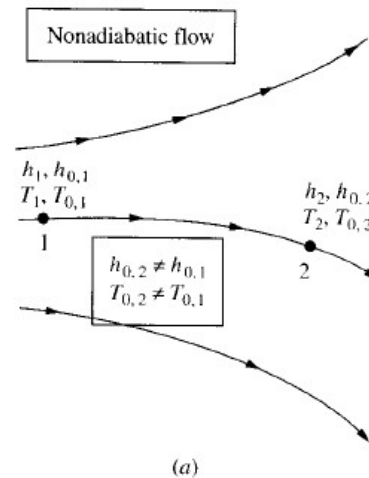
- The practical calculation of entropy is carried out as follows:
 - From 1st law of thermodynamics and enthalpy definition we have:

$$T ds = dh - vdp$$

How does this apply for a calorically perfect (ideal) gas?

Entropy balance and irreversibilities

- Adiabatic Process
- Reversible Process
- Isentropic Process
- How are the system properties related?



Why are the last equations so important?
Why are so frequently used?

Stagnation State

- We define the stagnation state of a gas as the state reached in decelerating a flow to rest reversibly and adiabatically and without any external work. Thus, the stagnation state is reached isentropically. This state is also referred to as the *total* state of the gas. The symbols for the stagnation state in this book use a subscript “t” for total. The total pressure is p_t , the total temperature is T_t , and the total density is ρ_t . Since the stagnation state is reached isentropically, the static and total entropy of the gas are the same, that is, $s_t = s$.
- For example

$$h_t \equiv h + \frac{V^2}{2}$$

What else can deduct from this equation?

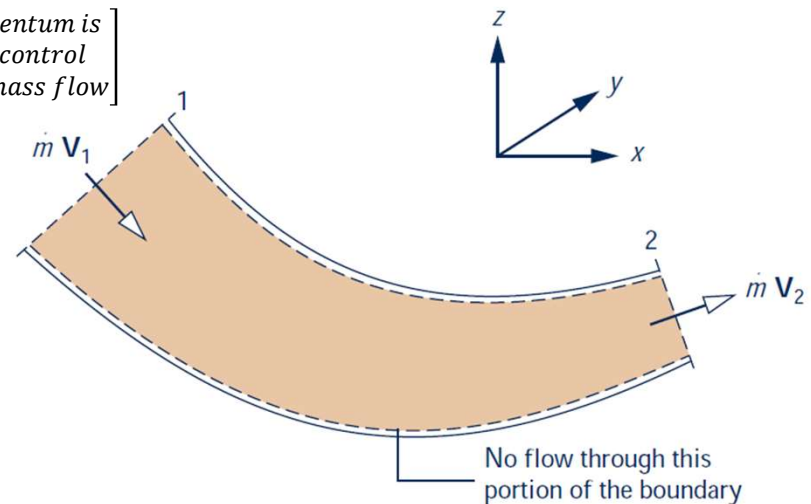
Momentum Equation

- The analysis of compressible flows requires the principles of conservation of mass and energy, the second law of thermodynamics, and relations among the thermodynamic properties of the flowing gas. In addition, Newton's second law of motion is required.

$$F = \dot{m}\vec{V} = \left[\begin{array}{c} \text{time rate of momentum} \\ \text{transfer into or} \\ \text{out of a control volume} \\ \text{accompanying mass flow} \end{array} \right]$$

$$\left[\begin{array}{c} \text{time of rate of change} \\ \text{of momentum contained} \\ \text{within the control volume} \end{array} \right] = \left[\begin{array}{c} \text{resultant force} \\ \text{acting on the} \\ \text{control volume} \end{array} \right] + \left[\begin{array}{c} \text{net rate at which momentum is} \\ \text{transferred into the control} \\ \text{volume accompanying mass flow} \end{array} \right]$$

$$F = \dot{m}_2 \vec{V}_2 - \dot{m}_1 \vec{V}_1 = \dot{m}(\vec{V}_2 - \vec{V}_1)$$



Momentum Equation

- Assuming that the frictional effects are negligible and thus, there are no shear forces and no external body forces in the flow, the momentum equation becomes a balance between static pressure forces and momentum transfer.

$$P_1 A_1 - P_2 A_2 = \dot{m}_2 \vec{V}_2 - \dot{m}_1 \vec{V}_1$$

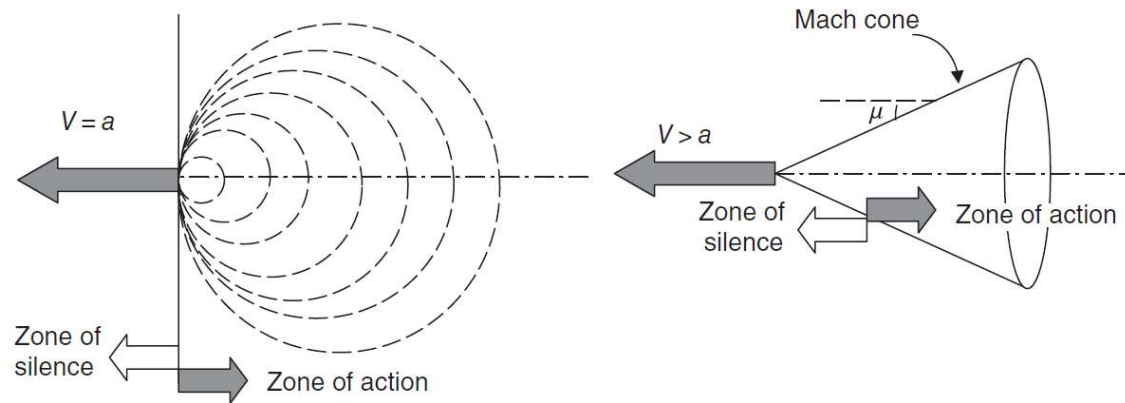
$$P_1 A_1 - P_2 A_2 = \dot{m}_2 \vec{V}_2 - \dot{m}_1 \vec{V}_1$$

$$\dot{m}_1 \vec{V}_1 + P_1 A_1 = \dot{m}_2 \vec{V}_2 + P_2 A_2$$

$$I_1 = I_2 \quad \leftarrow \text{Fluid impulse}$$

Speed of Sound & Mach Number

- The speed of sound is a local parameter, it is defined as the speed at which an infinitesimally small pressure wave travels through a medium. Its value depends directly on the gas temperature; hence it drops when fluid accelerates (or expands) and increases when gas decelerates (or compresses).



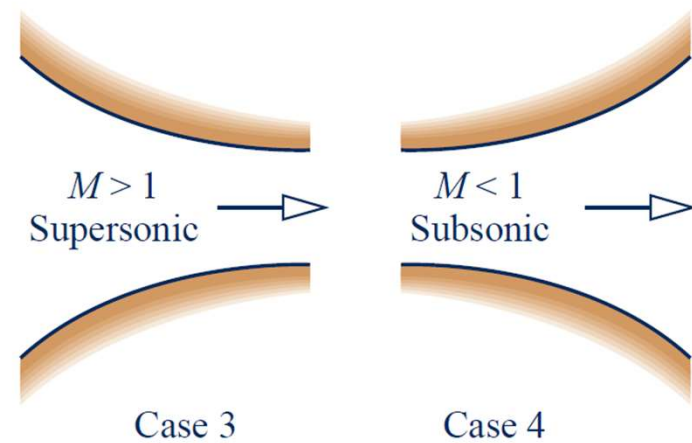
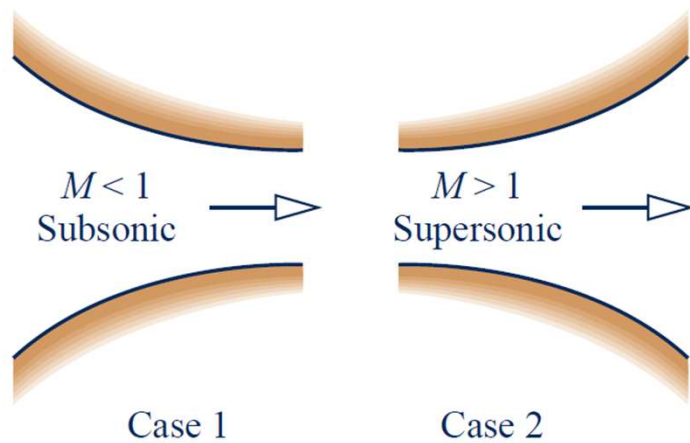
Compressible Flow through Nozzles and Diffusers

- From

$$\frac{dA}{A} = -\frac{dV}{V}(1 - M^2)$$

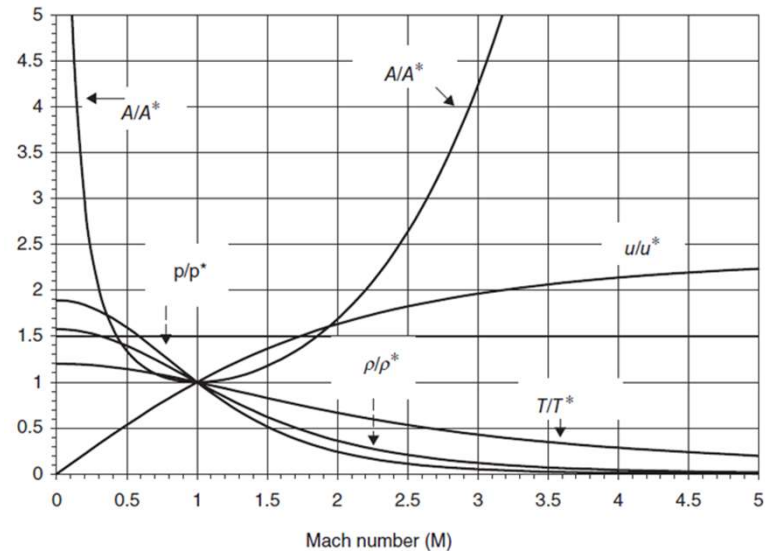
- We can observe 4 different scenarios for the variation of area and velocity
 - Case 1: Subsonic Nozzle. $dV > 0$, $M < 1 \Rightarrow dA < 0$: The duct *converges* in the direction of the flow
 - Case 2: Supersonic Nozzle. $dV > 0$, $M > 1 \Rightarrow dA > 0$: The duct *diverges* in the direction of the flow
 - Case 3: Subsonic Diffuser. $dV < 0$, $M > 1 \Rightarrow dA < 0$: The duct *converges* in the direction of the flow
 - Case 4: Supersonic Diffuser. $dV < 0$, $M < 1 \Rightarrow dA > 0$: The duct *diverges* in the direction of the flow

Compressible Flow through Nozzles and Diffusers



Sonic Throat

- The sonic state is where the gas speed and the local speed of sound are equal, it is distinguished with an asterisk.
- For example, the pressure at the sonic point is p^* and the temperature is T^* , the speed of sound is a^* , and so on.



Isentropic Relations as a function of Mach number in the Sonic Throat

Temperature

$$\frac{t}{t^*} = \frac{\frac{\gamma+1}{2}}{1 + \frac{\gamma-1}{2}M^2}$$

Area

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{1 + \frac{\gamma-1}{2}M^2}{\frac{\gamma+1}{2}} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

Pressure

$$\frac{p}{p^*} = \left(\frac{\frac{\gamma+1}{2}}{1 + \frac{\gamma-1}{2}M^2} \right)^{\frac{\gamma}{\gamma-1}}$$

Density

$$\frac{\rho}{\rho^*} = \frac{\frac{p}{p^*}}{\frac{t}{t^*}} = \left(\frac{\frac{\gamma+1}{2}}{1 + \frac{\gamma-1}{2}M^2} \right)^{\frac{1}{\gamma-1}}$$

Mass Flow at the Sonic Throat

- For given stagnation conditions, the maximum possible mass flow passes through a duct when its throat is at the critical or sonic condition. The duct is then said to be *choked* and can carry no additional mass flow unless the throat is widened. If the throat is constricted further, the mass flow through the duct must decrease.

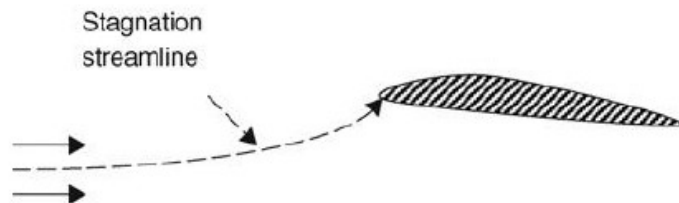
$$\begin{aligned} \dot{m} &= \rho V A \\ \downarrow \\ NDMF &= \frac{\dot{m} \sqrt{T_t}}{P_t A} = M \sqrt{\frac{\gamma}{R}} \left(\frac{1}{1 + \frac{\gamma-1}{2} M^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \end{aligned}$$

Some Aspects of Supersonic Flows: Shock Waves

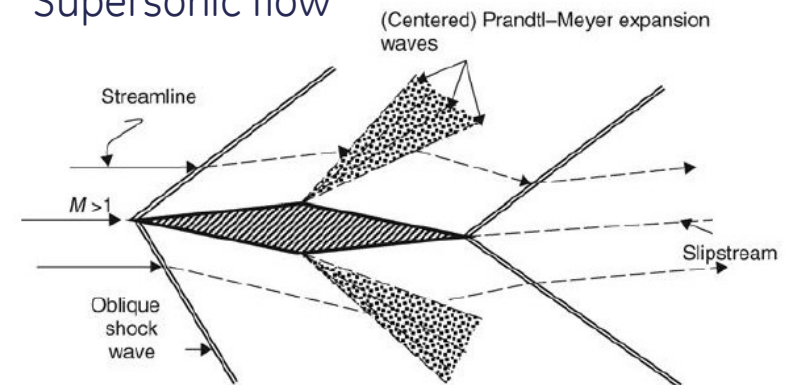
What is a shockwave?

- A large-amplitude compression wave, such as that produced by an explosion, caused by supersonic motion of a body in a medium.

Subsonic flow

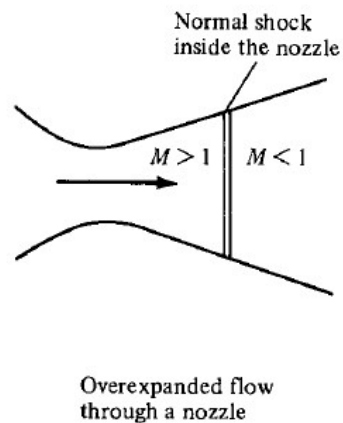
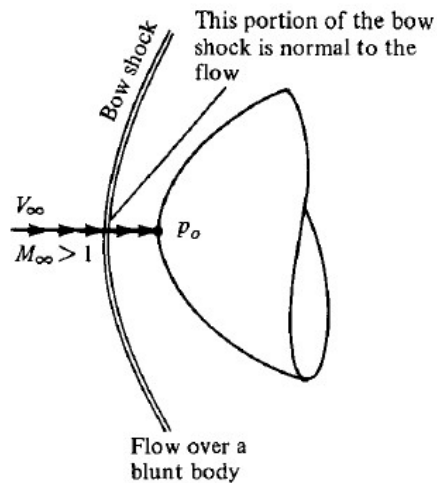


Supersonic flow



Normal Shockwaves

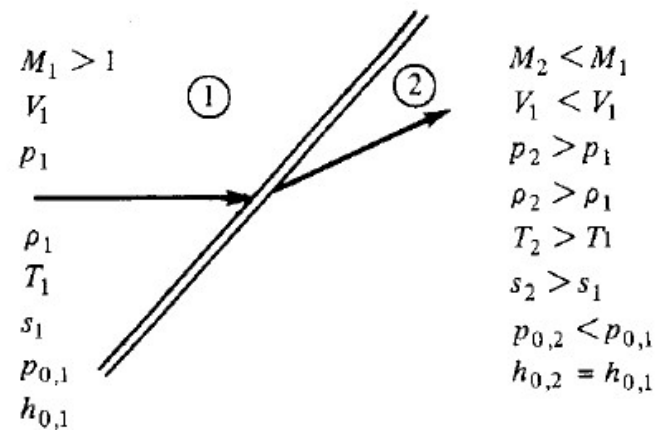
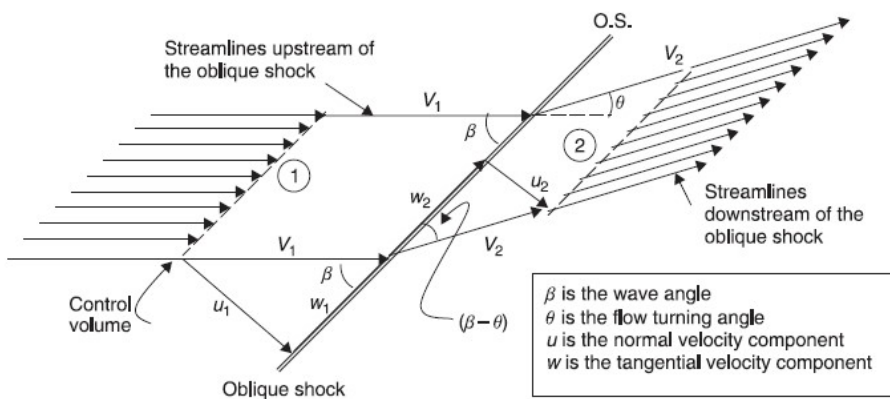
Where are these shockwaves normally present?



How properties change across the shockwave?

$M_1 > 1$		$M_2 < 1$
V_1	①	$V_2 < V_1$
p_1		$p_2 > p_1$
ρ_1		$\rho_2 > \rho_1$
T_1		$T_2 > T_1$
s_1		$s_2 > s_1$
$p_{0,1}$		$p_{0,2} < p_{0,1}$
$h_{0,1}$		$h_{0,2} = h_{0,1}$

Oblique Shockwaves



Dimensionless Parameters

- Following the Buckingham Π theorem, a group of independent dimensionless parameters appear in fluid mechanics that govern its behavior and hence introduce dynamic similarity between flows that share these parameters. The major dimensionless groups that are relevant to our studies are
 - Mach Number
 - Reynolds Number
 - Prandtl Number
 - Nusselt Number
 - Stanton Number
 - Pressure Coefficient
 - Force Coefficient
 - Knudsen Number